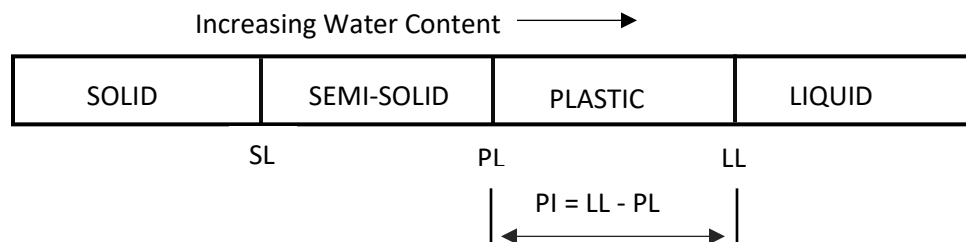


Experiment	Atterberg Limits Test
Objective	To determine the liquid limit, plastic limit and the plasticity index of a fine-grained soil sample.
Apparatus	Liquid limit device, grooving tool, balance readable to 0.01g Porcelain dish, glass plate, 425µm sieve, spatula Wash bottle, moisture can, drying oven
Theory	The Atterberg limits are used to identify fine-grained soils like silts and clays. Depending on the water content, a soil may appear in one of the four consistency states: solid, semi-solid, plastic and liquid. In each state, consistency and behavior of a soil is different and consequently so are its engineering properties.



The scientist Atterberg first defined the boundary between each consistency state based on a change in the soil's water content. The Liquid Limit(LL), Plastic Limit(PL) and Shrinkage Limit (SL) define the water content boundaries between liquid, plastic, semi-solid and solid states. The plasticity index (PI) defines the complete range of plastic state.

Liquid limit is the water content at which soil changes from a plastic to a liquid state when the soil specimen is just fluid enough for a groove to close under test conditions.

Plastic limit is the water content at the change from a plastic to a semi-solid state. This test involves repeatedly rolling a soil sample into a thread until it reaches a point where it crumbles.

Shrinkage limit is the water content where the further loss of moisture does not cause a decrease in specimen volume.

Procedure

Liquid limit test

1. Air-dry the soil sample and break the clods. Remove the organic matter.
2. Take about 300g of the specimen passing through 425 μ m sieve and mix thoroughly with distilled water in the porcelain dish.
3. Place a portion of the paste in the cup of the liquid limit device and level the mix so as to have a maximum depth of 1cm.
4. Draw the grooving tool through the sample along the symmetrical axis of the cup, holding the tool perpendicular to the cup.
5. After the soil pat has been cut, rotate the handle at the rate of 2 blows per second and count the number of blows, till the two parts of the soil sample come into contact along a 13mm length.
6. Take about 10g of soil near the closed groove and determine its water content.
7. Transfer the soil of the cup to the dish and repeating the test, obtain at least 5 readings in the range of 15 to 50 blows.

Plastic limit test

1. Take about 8g of the prepared soil and roll it with fingers on a glass plate.
2. If the diameter of the threads can be reduced to less than 3mm, without any cracks appearing, it means that the water content is more than its plastic limit. Knead the soil to reduce the water content and roll it into a thread again.
3. Repeat the process of alternate rolling and kneading until the thread crumbles at a diameter of 3 mm.
4. Collect and keep the pieces of crumbled soil thread in a container to determine the water content.
5. Repeat the process at least twice more with fresh samples of soil each time.

Calculations

Calculate the water content for each specimen.

LL – Plot on a semi-log graph, with no of blows in x axis (log scale) and the water content in y axis and draw the best straight line through the plotted points. The water content corresponding to 25 blows is taken as the LL of the soil.

PL– The average water content of the crumbled threads is taken as the PL of the soil.

Calculate the PI value.

Results

Liquid Limit, Plastic Limit and the Plasticity Index of the soil sample

Conclusion

Identify the soil according to the Atterberg limits.

Observations

Atterberg Limits Test

Name -----

Reg Number -----

Course -----

Date -----

Liquid Limit Test

Trial No	Moisture can No	Mass of can (g)	Mass of wet soil + can (g)	Mass of dry soil + can (g)	No of Blows
1					
2					
3					
4					
5					

Plastic Limit Test

Trial No	Moisture can No	Mass of can (g)	Mass of wet soil + can (g)	Mass of dry soil + can (g)
1				
2				
3				